

1) a) $x > 0$: $x = 0$: $x < 0$:

$-kx \leftarrow \boxed{\quad} \quad \boxed{\quad} \quad \boxed{\quad} \rightarrow kx$

b) $F = ma \rightarrow -kx = m\ddot{x} \rightarrow m\ddot{x} + kx = 0 \rightarrow \ddot{x} + \frac{k}{m}x = 0 \rightarrow \ddot{x} + \omega_0^2 x = 0$

c) $x(t) = e^{\lambda t} \quad \dot{x}(t) = \lambda e^{\lambda t} \quad \ddot{x}(t) = \lambda^2 e^{\lambda t}$

$\lambda^2 e^{\lambda t} + \omega_0^2 e^{\lambda t} = 0 \rightarrow \lambda^2 = -\omega_0^2 \rightarrow \lambda_{1,2} = \pm i\omega_0$

$x(t) = A e^{-i\omega_0 t} + B e^{+i\omega_0 t}$

d) $x(0) = C = x_0 \rightarrow x(t) = x_0 \cos(\omega_0 t)$

e) $F = ma \rightarrow -kx - b\dot{x} = m\ddot{x} \rightarrow m\ddot{x} + b\dot{x} + kx = 0 \rightarrow \ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = 0$

$\Rightarrow \ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$

f) $\frac{1}{2} = \frac{x_0 e^{-\beta t}}{x_0} \rightarrow e^{\beta t} = 2 \rightarrow \ln e^{\beta t} = \ln 2 \rightarrow \beta t = \ln 2 \rightarrow t = \ln 2 / \beta$

$t = \frac{2m \ln 2}{b} = \frac{2 \times 1 \times 0.7}{1.4} \frac{\text{kg}}{\text{kg}} \text{s} = 1 \text{s}$

g) $\omega_1 = \sqrt{\omega_0^2 - \beta^2} = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} = \sqrt{\frac{1.3}{1} \frac{\text{kg}}{\text{s}^2 \text{kg}} - \frac{1.4 \times 1.4}{4 \times 1^2} \frac{\text{kg}^2}{\text{s}^2 \text{kg}^2}}$
 $= \sqrt{1.3 \frac{1}{\text{s}^2} - 0.7 \times 0.7 \frac{1}{\text{s}^2}} = \sqrt{1.3 - 0.49} \frac{\text{rad}}{\text{s}} = \sqrt{0.81} \frac{\text{rad}}{\text{s}} = 0.9 \frac{\text{rad}}{\text{s}}$

$\omega_1 = 2\pi f_1 \rightarrow f_1 = \omega_1 / 2\pi = \frac{0.9}{2 \times 3.14} \text{ Hz} = 0.15 \text{ Hz} \quad 0.15 \text{ times!}$

$$2) a) \bar{F}_g = mg = \rho_w \frac{4}{3} \pi R^3 g$$

$$b) D = mg$$

$$c) \frac{1}{2} C \rho_a A v_t^2 = \rho_w \frac{4}{3} \pi R^3 g \rightarrow \frac{1}{2} C \rho_a \pi R^2 v_t^2 = \rho_w \frac{4}{3} \pi R^3 g$$

$$C \rho_a v_t^2 = \rho_w \frac{8}{3} R g \rightarrow v_t = \sqrt{\frac{8}{3} \frac{\rho_w}{\rho_a} \frac{R g}{C}}$$

$$d) v_t = \sqrt{\frac{\cancel{8}}{\cancel{3}} \frac{1000}{1.0} \frac{\cancel{1.5 \times 10^{-3}} \times 10}{\cancel{0.80}} \frac{m}{s^2}} = \sqrt{\frac{10 \frac{m^2}{s^2}}{0.10 s^2}} = \sqrt{100 \frac{m^2}{s^2}} = 10 \frac{m}{s}$$

$$10 \frac{m}{s} \times \frac{1 km}{1000 m} \times \frac{3600 s}{1 h} = 10 \times 3.6 \frac{km}{h} = 36 \frac{km}{h} : \text{not faster than the speed limit in Prague.}$$

$$e) V = \sqrt{2 \times 10 \times 1000 \frac{m}{s^2}} = \sqrt{2 \sqrt{10000}} \frac{m}{s} = \sqrt{2} \times 100 \frac{m}{s} \approx 140 \frac{m}{s}$$

$$140 \frac{m}{s} \times \frac{1 km}{1000 m} \times \frac{3600 s}{1 h} = 140 \times 3.6 \frac{km}{h} = 36 \times 14 \frac{km}{h} = 504 \frac{km}{h} : \text{as fast as a jet plane!}$$

$$3) a) mgh = \frac{1}{2} k x_m^2 \rightarrow mg l_1 \sin \theta = \frac{1}{2} k x_m^2 \rightarrow \sqrt{\frac{2mg l_1 \sin \theta}{k}} = x_m$$

b) Yes, it can. No friction, no mechanical energy loss:

$$mgh \rightarrow \frac{1}{2} m v^2 \rightarrow \frac{1}{2} k x_m^2 \rightarrow \frac{1}{2} m v^2 \rightarrow mgh$$

$$c) F_f = \mu_k N = \mu_k mg$$

$$d) \vec{F}_f \leftarrow \boxed{\text{block}} \rightarrow d\vec{r} \quad W = \int \vec{F}_f \cdot d\vec{r} = -F_f l_2 = -\mu_k mg l_2$$

$$e) mg l_1 \sin \theta - \mu_k mg l_2 = \frac{1}{2} k x_m^2 + \mu_k mg x_m$$

$$\textcircled{\frac{1}{2} k x_m^2}^A + \textcircled{\mu_k mg x_m}^B + \textcircled{\mu_k mg l_2 - mg l_1 \sin \theta}^C = 0$$

$$x_{1,2} = \frac{-\mu_k mg \pm \sqrt{\mu_k^2 m^2 g^2 - 2 \frac{1}{2} k (\mu_k mg l_2 - mg l_1 \sin \theta)}}{k}$$

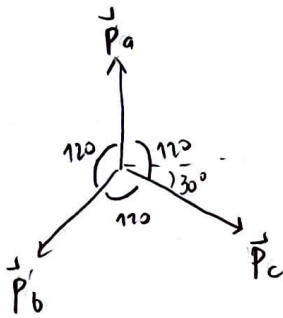
f) No, it can't. Some mechanical energy is lost during the block covers the distance l_2 .

$$mgh \rightarrow \frac{1}{2} m v^2 \rightarrow \frac{1}{2} k x_m^2 \rightarrow \frac{1}{2} m v^2 \rightarrow mgh' \text{ where } h' < h.$$

$\swarrow \quad \quad \quad \searrow \quad \quad \quad \swarrow$
 $\mu_k mg l_2 + \mu_k mg x_m \quad \quad \mu_k mg x_m \quad \quad \mu_k mg l_2$

$$\underline{\text{Total energy loss}} = 2\mu_k mg l_2 + 2\mu_k mg x_m = 2\mu_k mg (l_2 + x_m)$$

4) a)



$$b) \quad 0 = m V_c \cos 30^\circ - m V_b \cos 30^\circ$$

$$\Rightarrow V_b = V_c$$

$$c) \quad 0 = m V_a - m V_b \sin 30^\circ - m V_c \sin 30^\circ$$

$$V_a = \sin 30^\circ (V_b + V_c)$$

$$d) \quad \text{Let us say } V_b = V_c \equiv V$$

$$V_a = \frac{1}{2} 2V \equiv V \Rightarrow V_a = V_b = V_c = 2 \frac{m}{s}$$